「Def. Let V be a vector space,
Given a vector $\alpha \in V$ and linear operator $T : V \to V$, the subspace

$$Z(\alpha, T) \overset{\text{def}}{=} \{ g(T)\alpha \mid g(t) \in F[t] \} = \operatorname{span}\{ T^k \alpha \mid k \in \mathbb{N} \cup \{0\} \}.$$

is called the T-cyclic subspace generated by α

「Def. Let V be a vector space. Given $\alpha \in V$ and a linear operator T on V, an ideal

$$M(\alpha, T) = \{ g(t) \in F[t] \mid g(T)\alpha = 0 \}$$

is called the T-annihilator of α, and the unique monic generator of also called ".

「Thm. Let α be any non-zero vector in V and let p_α be the T-annihilator of α.
1) $\deg p_\alpha = \dim Z(\alpha, T)$
2) $Z(\alpha, T) = \operatorname{span}\{ T^k \alpha \mid 0 \leq k \leq \deg p_\alpha \}$
3) The minimal polynomial of $T|_{Z(\alpha, T)}$ is p_α.